PSU - DPT Curriculum Concepts

This working draft document provides explains concepts underlying the curriculum, particularly the concept-based component of the hybrid model, and the scaffolding and snowballing of the patient / client management course sequence.

Curriculum Concepts

Concepts within a curriculum are broad and abstract, universal in application and timeless.[1] While a conversation has started around concept-based curriculum in physical therapy there is still a need to define the "big ideas." As presented by Gwyer and Hack (2015) it is quite reasonable that the human movement system is an organizing principle from which to develop big ideas for the profession.[2] The concepts we have identified are broader and more abstract than the human movement system, but are considered through the focusing lens of the movement system in the curriculum. The rationale for broader and more abstract are to provide students with a framework for reasoning about and working with the concept of the movement system. We believe that DPT education has always had the detailed facts and a somewhat formulated concept of a movement system. What we are working toward now is a more fully developed systems theoretic understanding of the movement system.

Defined by the APTA:[3] "*The human movement system comprises the anatomic structures and physiologic functions that interact to move the body or its component parts."*

Systems
A system is a set of interacting or interdependent component parts forming a complex/intricate whole.[4] While physical therapists work with the movement system at different hierarchical scales (cellular inflammation and whole body movement during functional tasks in various environments), they also work within broader systems (ethics / values systems, psychosocial systems and health systems). Therefore it seems worthwhile to have a broader perspective on systems and to understand the interactions between various system components. Additionally we hope that a conceptual focus on systems will provide students with deeper insights into the human movement system - insights yet to be realized. And that combined with an understanding of adaptation and causation (both below) students will be better prepared for understanding, studying and applying empirical evidence to knowledge development and knowledge in practice.[5]

Adaptation
Something we know that systems do - including the human movement system - is change with time. These changes are often adaptations in response to internal or external disturbances. The study of systems includes a relatively new branch of investigation referred to as complex adaptive systems (also dynamic systems) to put emphasis on the adaptive process. Adaptation as a process does not necessarily involve a particular anatomic or physiologic function or interaction, but a process made possible by changes to system components and interactions. There is a hierarchy of adaptations that corresponds to the hierarchy of systems and the hierarchy of system components (genetic, epigenetic, biological, physiological, anatomical, behavioral, cultural / social). Courses that emphasize adaptation (particularly the Dynamic Systems sequence) pull all concepts on adaptive processes together that would have been covered in separate courses (such as anatomical changes, healing, physiological adaptations such as to immobility and exercise, motor learning) so that students can better understand the

underlying processes as well as the interactions and similarities.

We believe much of what physical therapists do is attempt to curtail adaptations across the hierarchy that have a negative impact on life (including movement), and promote adaptations that have positive impact on life. Emerging beyond the individual, the APTA's current Vision statement explicitly calls for a "transformation of society" which is essentially an adaptation of culture. We hope that an understanding of systems and system adaptation (dynamic systems, complex adaptive systems) helps prepare graduates for innovation and transformation. Being able to see not only how things are with better understanding of complex adaptive interactions, but see how things could be improved through adaptive processes.

Causation

Even more general than systems and their adaptation is the study of causation. For any attempt to understand, predict future states or manipulate a system and/or its adaptation we must understand causation. Cause - effect relationships are the foundational concept of all scientific inquiry and of all practice. The only reason we believe our treatment will work is that we believe there is a cascade of mechanisms triggered by the treatment. The only reason we believe we can diagnosis based on signs and symptoms is because we believe certain conditions cause a certain rather unique pattern of signs and symptoms.

Understanding causation can seem rather simple really. What is complicated is understanding causation in a real world open complex system where there are many components and a vast hierarchy of emerging and interacting parts. There is a stark contrast between closed and open systems, particularly when it comes to the study of causation within these systems. This contrast underlies the classic problem of generalization - how to take an understanding about causal relationships from a closed system (experimental conditions) to the patient in an open system (real life). Therefore the inclusion of causation as a concept is primarily focused on attempting to understand the ways of knowing, bias and learning causal relationships in light of this underlying complex interaction. This includes the consideration of graphical causal model representations of causal networks (as logical, mathematical structures) that then allow the development of reasoning patterns based on the network structure. This is an overarching approach for both how knowledge is developed from evidence (evidence based practice), and how we should practice based on the knowledge (knowledge based practice).[6]

Goals

We hope these concepts will provide a framework for the concept of movement. They aim to provide provide students a framework for:

1. How to reason and develop knowledge about movement generally (causation and inference)
2. How systems and their hierarchical structure:
   - emerge (movement emerges from the movement system which emerges from constituent parts)
   - provide support and capabilities for our experiences (i.e. dynamic stability, control and feedback)
   - interact (humans interact and emerge communities, which interact and emerge as society)
3. How systems adapt (hierarchy of adaptation and adaptability of movement, environments, communities and societies)

These concepts are primarily developed through the theory courses in the curriculum but form the dependencies upon which the much of the practice courses are based. The sole purpose of the theory courses is to infuse conceptual understanding into the practice of physical therapy that is covered in the practice courses (Table 1, note Theory course clusters are explained in the next section). The practice courses are a scaffold that snowballs through the curriculum. This snowball gathers mass both from new practice content, but also from greater theoretical conceptual understanding.

Table 1. Theory Course Relationship to Concepts

Theory Course Cluster	Causation	Systems Theory	Adaptation
Knowledge Foundations	Existing knowledge about causal relationships; represented as causal networks	Recognition that these foundations are progressively embedded in a complex system, governed by causation but making causation more difficult to detect	Existing knowledge adaptation; includes what causes adaptation, and how system interactions result in adaptation (complex adaptive system)
Systems Theory	Considers how system structure and causal structure relate, use system models as a starting point for reasoning about causal models (and networks)	Abstract principles underlying systems (interaction of parts and their interconnections that come together for a purpose) including explicit models, underlying assumptions, initial conditions, non linearity, regulation, emergence	Dynamic and complex adaptive systems; stability, stationarity, stasis and equilibrium; includes hierarchy of adaptation, healing, motor learning, development and growth
Knowledge Development	Empirical observations to causal knowledge; rational development of causal models (networks and structure); causal assumptions and bias; inferential reasoning	Confounding, effect modification; development and appraisal of knowledge and therefore guidelines for practice	Develop an appreciation for detecting adaptive changes, particularly when attempting to predict changes that will occur (prognosis); diagnosis the cause of changes; or prescribe interventions to induce adaptation

Theory & Practice

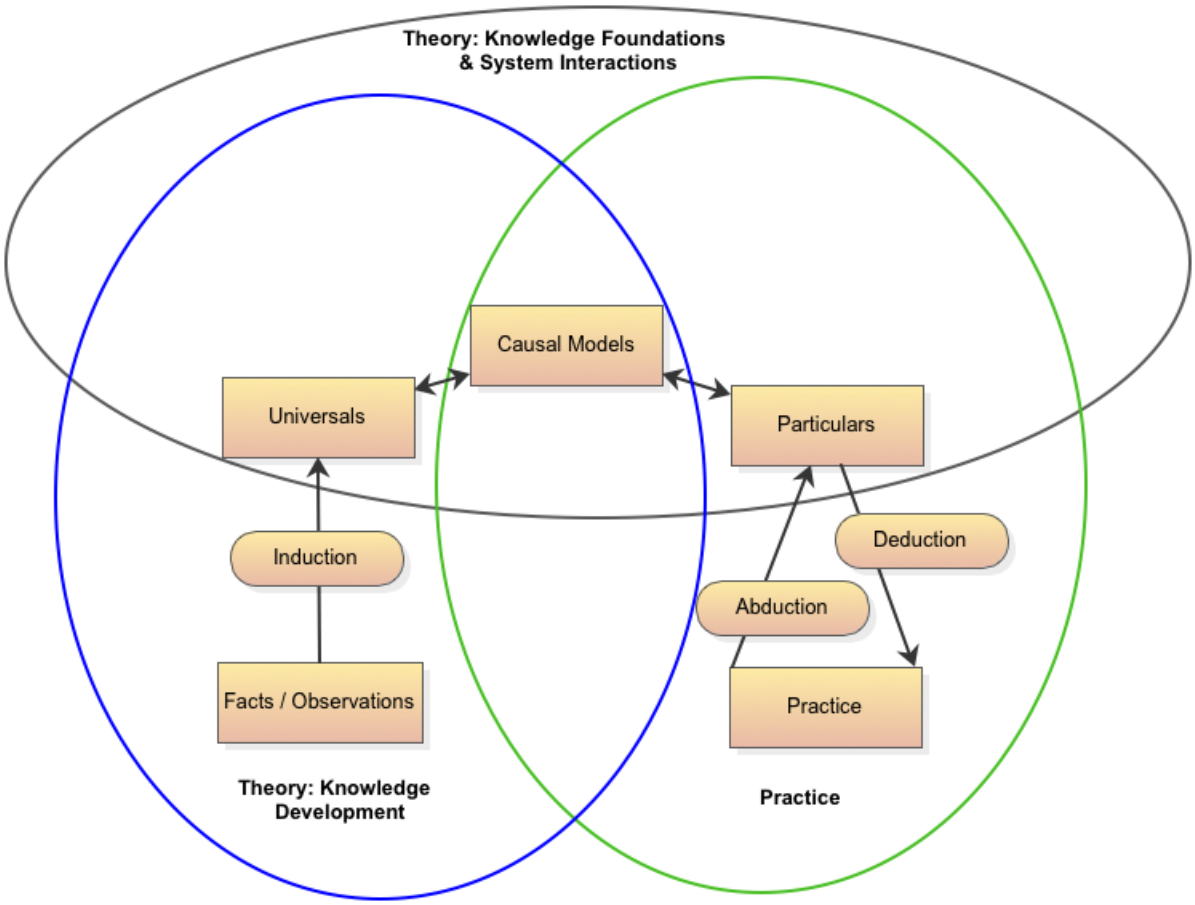
Theory & Practice is a emphasized through the curriculum. There is no question that graduates are being prepared for the professional practice of physical therapy. We view practicing physical therapy as a DPT professional as a "whole complex task." Preparation to practice requires a certain standard of behavior, knowledge, skill and attributes based on contemporary practice knowledge, expectations, ethics, values and legal guidelines. It is "complex learning" and therefore "involves integration of these knowledge, skills and attitudes; coordinating qualitatively different constituent skills, and often transferring of what is learned in the school or training setting to daily life and work settings."[7] Approximately half of the course credit and more than half of the time in the DPT program is dedicated fully to practice based coursework (whole complex task learning) both on campus through courses, labs and integrated clinical; and off campus through clinical education experiences.

In addition to practice courses integrating "whole task complexity", there are two other components to the practice courses. One is "procedural information" which is provided through labs, simulation and clinic practice (also could be considered the psychomotor domain of learning), as well as "part-task practice" which is provided in labs, lab practice and curricular repetition which is focused on development of the automatic psychomotor and cognitive capabilities of students. The goal being practice of commonly utilized skills with minimal cognitive demand so that - like a more experienced therapist - they can be thinking a few steps ahead during patient / client management activities.

Students learn through the concepts of causation, systems theory and adaptation that knowledge from which practice is based ultimately derives from an understanding of a vast system of causal relationships built up from both structured and unstructured observations, from which emerge universals which then are organized as a causal network. Causal networks are represented as causal models. Knowledge based practice is practice based on the knowledge represented by a set of causal models. This representative knowledge is provided through theory courses as what Van Merrienboer and Kirschner (2012) call "supplemental information."

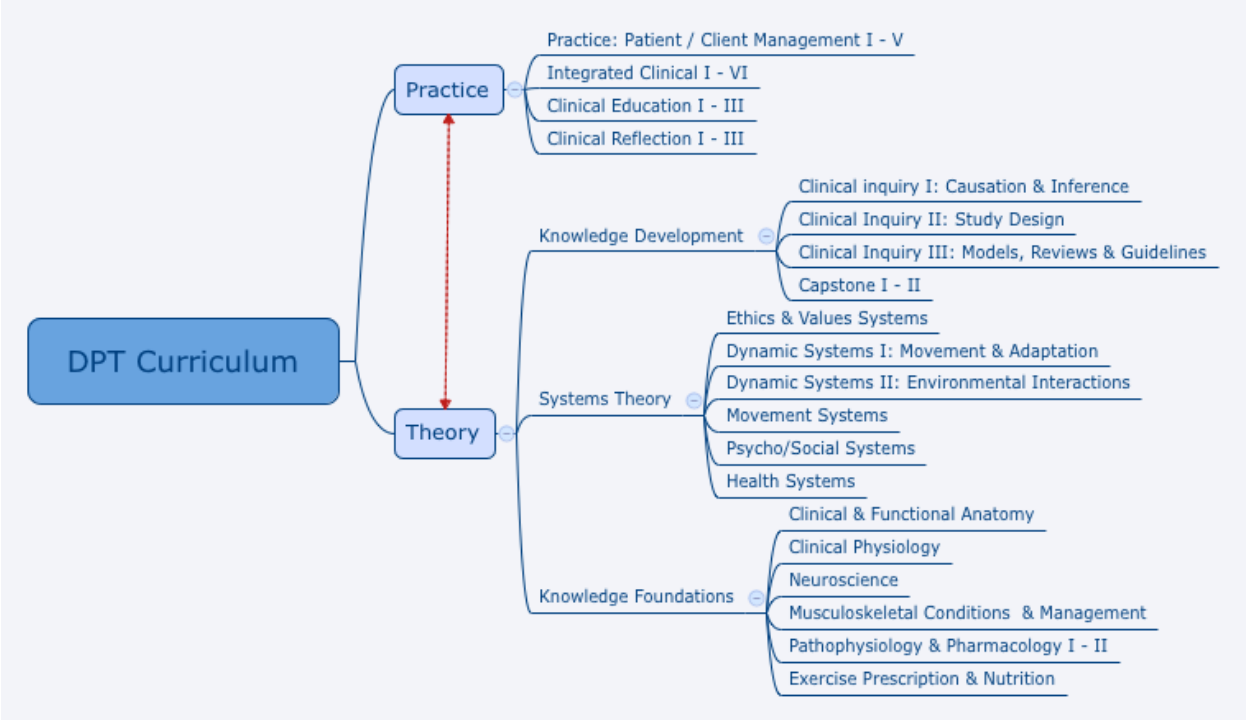
The Theory courses are divided into three clusters: Knowledge Development, Knowledge Foundations and System Interactions. Figure 1 depicts the knowledge development process where universals emerge from the empirical observation of facts and observations (left side of the graphic). This universal knowledge is then embedded into a causal model. The causal model represents an understanding of causal system structure, its components and interactions. Practice proceeds downward from knowledge about causal system structure to particular situations (patient, client, community) to practice decisions (decisions the describe, proscribe or prescribe explicit actions). Figure 2 provides a graphic of the curriculum theory and practice courses.

Figure 1. Theory & Practice



Throughout the curriculum there are several connections between theory and practice built explicitly into courses. The Clinical Reflection course connects practice to knowledge through practical reflection, and the Capstone requires critical analysis of existing knowledge and knowledge development processes of current clinical practice, including recommendations and guidelines.

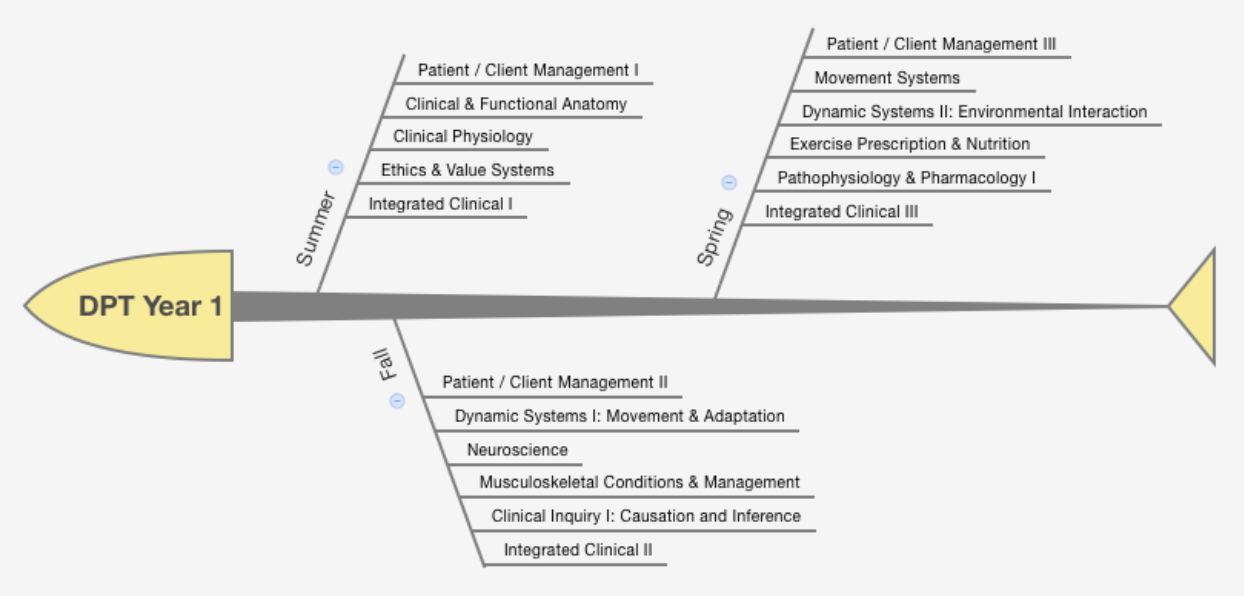
Figure 2. Graphic of Curriculum Theory & Practice Courses

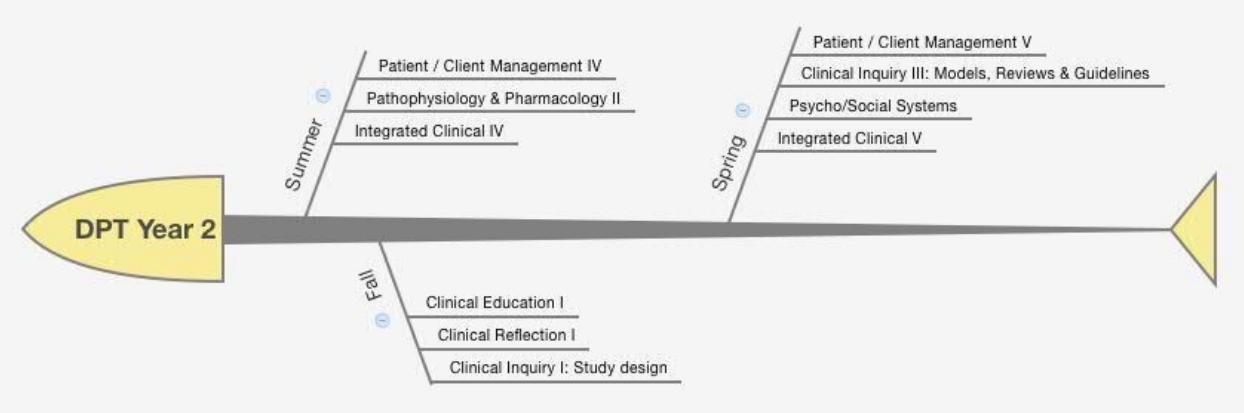


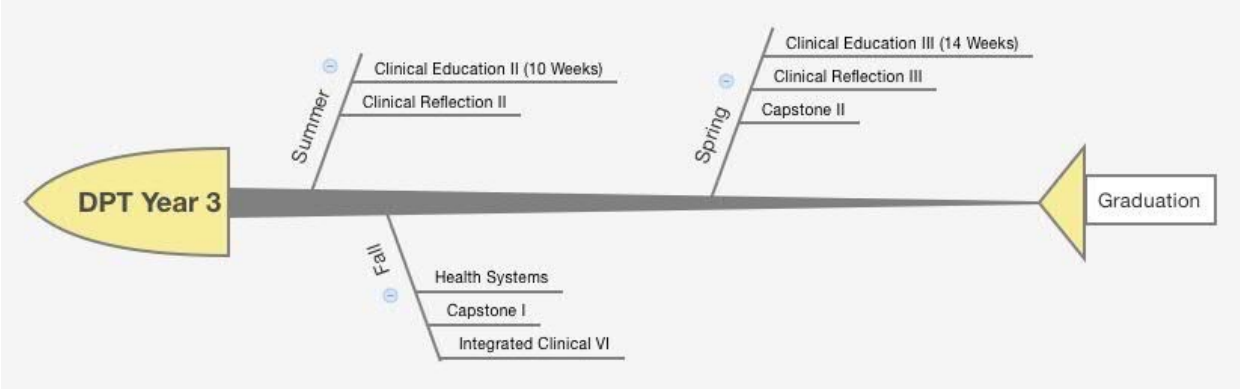
Practice Integration and Unification

The practice course sequence consists of the Patient / Client Management sequence (PCM 1 - 5) with increasing credits and therefore time for the course during the sequence. PCM 1 provides an overview of patient / client management (scaffold, "whole" picture) from which to build on through the curriculum (snowballing). This scaffold is not just for materials covered in PCM, but also for conceptual knowledge (supporting information) developed in theory courses. Through the curriculum there is a decrease in the number of total courses and an increase in the number of practice course credit in light of the practice orientation of becoming a Doctor of Physical Therapy (Figure 3). There is integration (to avoid compartmentalization of knowledge) and unification (to avoid fragmentation of knowledge) of concepts and the complexity of problems to be solved as the student progresses toward entry level practice.

Figure 3. Graphic of Curricular Sequence



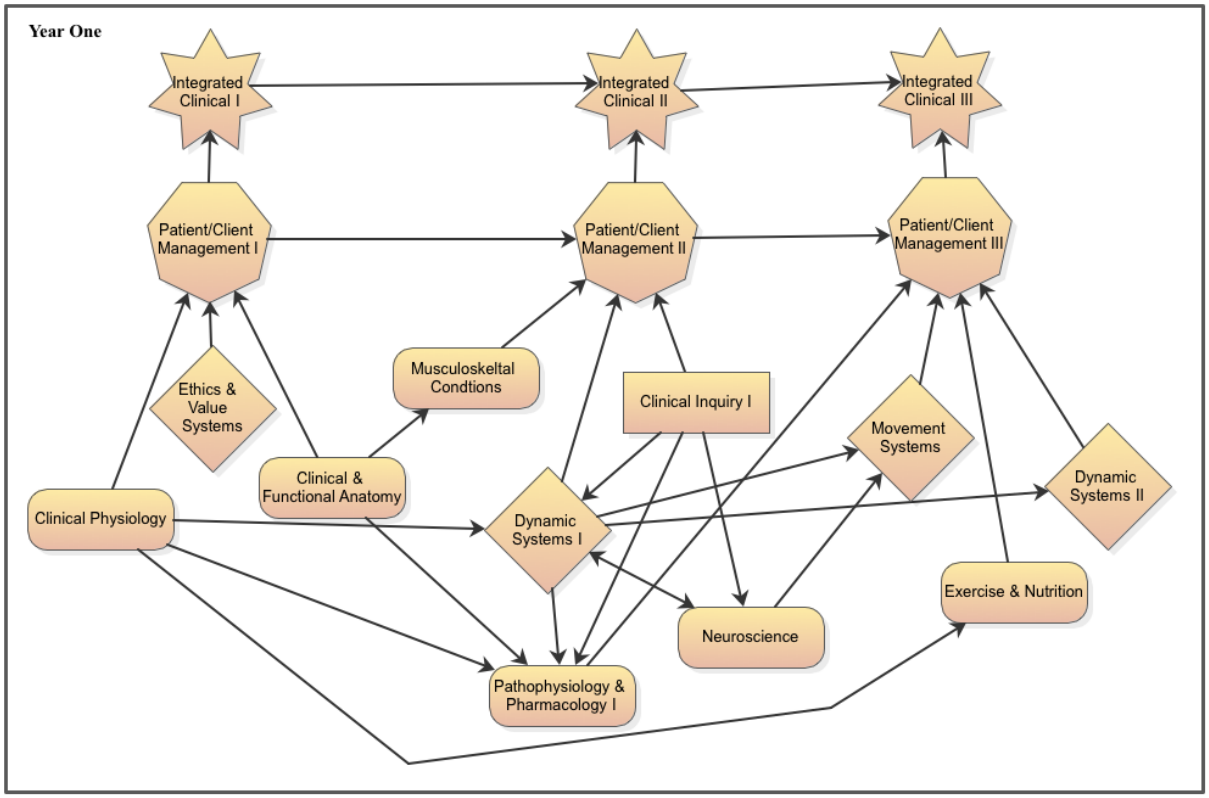




Curricular Interactions

Through the curriculum there are interactions between the courses that form the basis for the integration of concepts and supplemental information into practice and contribute to the snowballing of the initial scaffold developed for patient / client management. Figure 4 attempts to capture the primary course interactions as a sketch of the curricular system from which the course dependencies (hard dependencies are prerequisites and soft dependencies are co-requisites) are based.

Figure 4. Graphic of Curricular Interactions & Flow



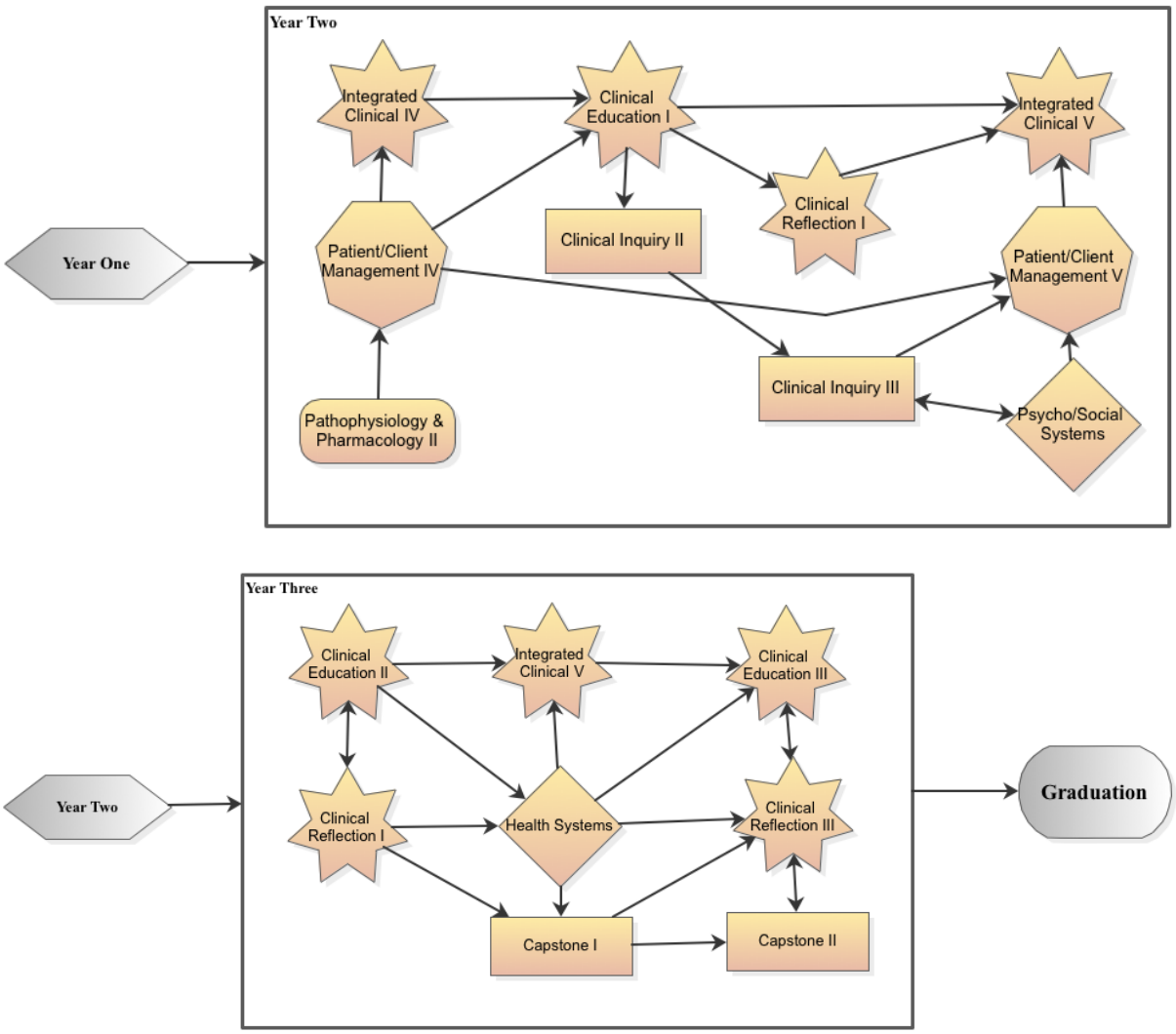

Year Two
Integrated Clinical IV
Clinical Education I
Integrated Clinical V
Clinical Reflection I
Year One
Patient/Client Management IV
Clinical Inquiry II
Patient/Client Management V
Clinical Inquiry III
Pathophysiology & Pharmacology II
Psycho/Social Systems
Year Three
Clinical Education II
Integrated Clinical V
Clinical Education III
Year Two
Clinical Reflection I
Health Systems
Clinical Reflection III
Graduation
Capstone I
Capstone II

Figure 5.

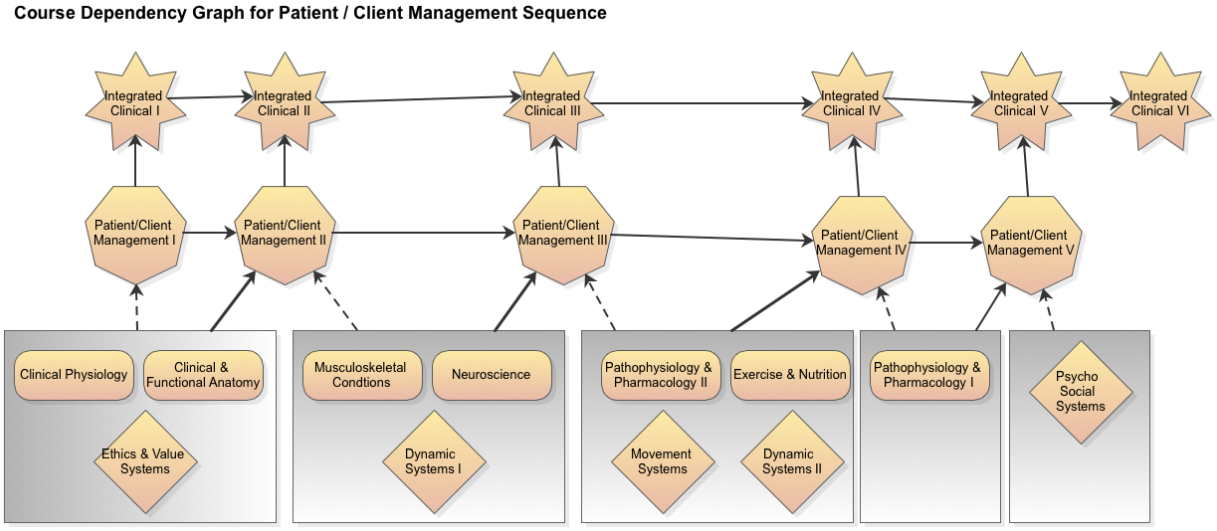


Figure 5 is a simplified (clustered) version with a focus on the patient / client management sequence (dotted lines indicate soft dependency and solid lines indicate hard dependency).

Figures 6 - 12

Figures 6 - 11 depict the scaffold of patient / client management and then the explicit courses that form the hard (left column) and soft (right column) dependencies.

Figure 12 provides slightly more insight into the content flow as the sequence snowballs. This graphic only includes the hard dependencies (pre requisite courses that precede the PCM course).

Figure 6. Patient / Client Management

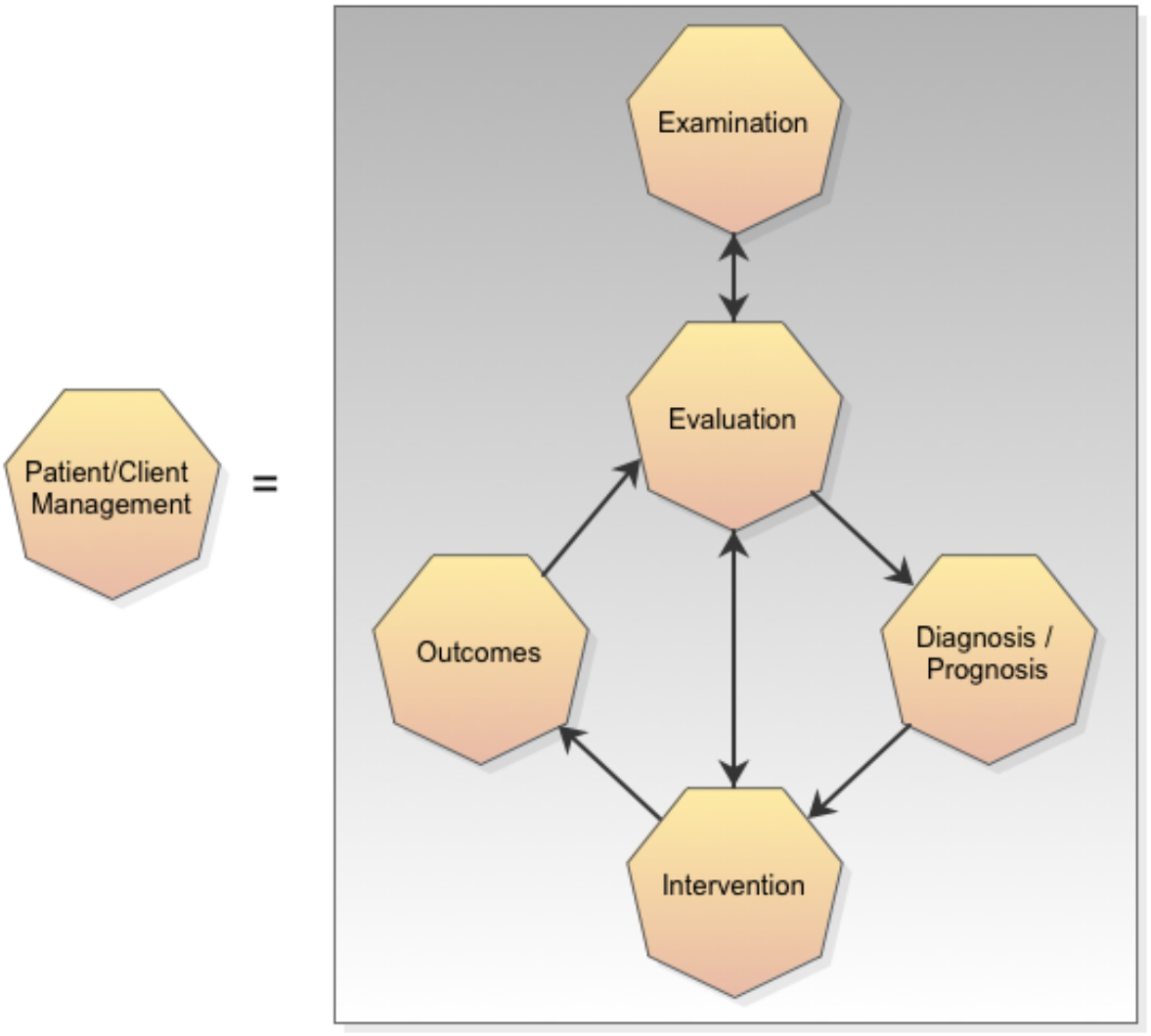
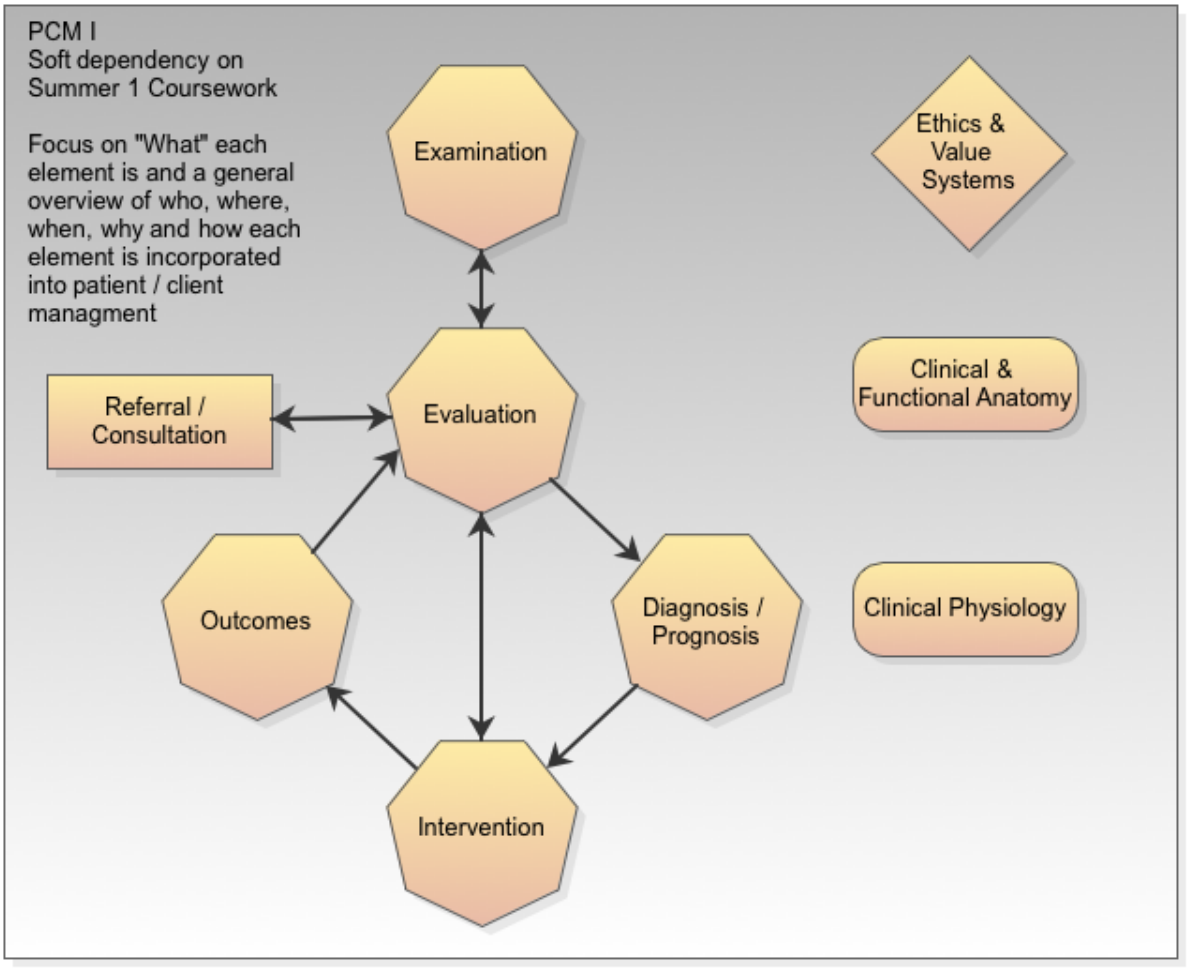
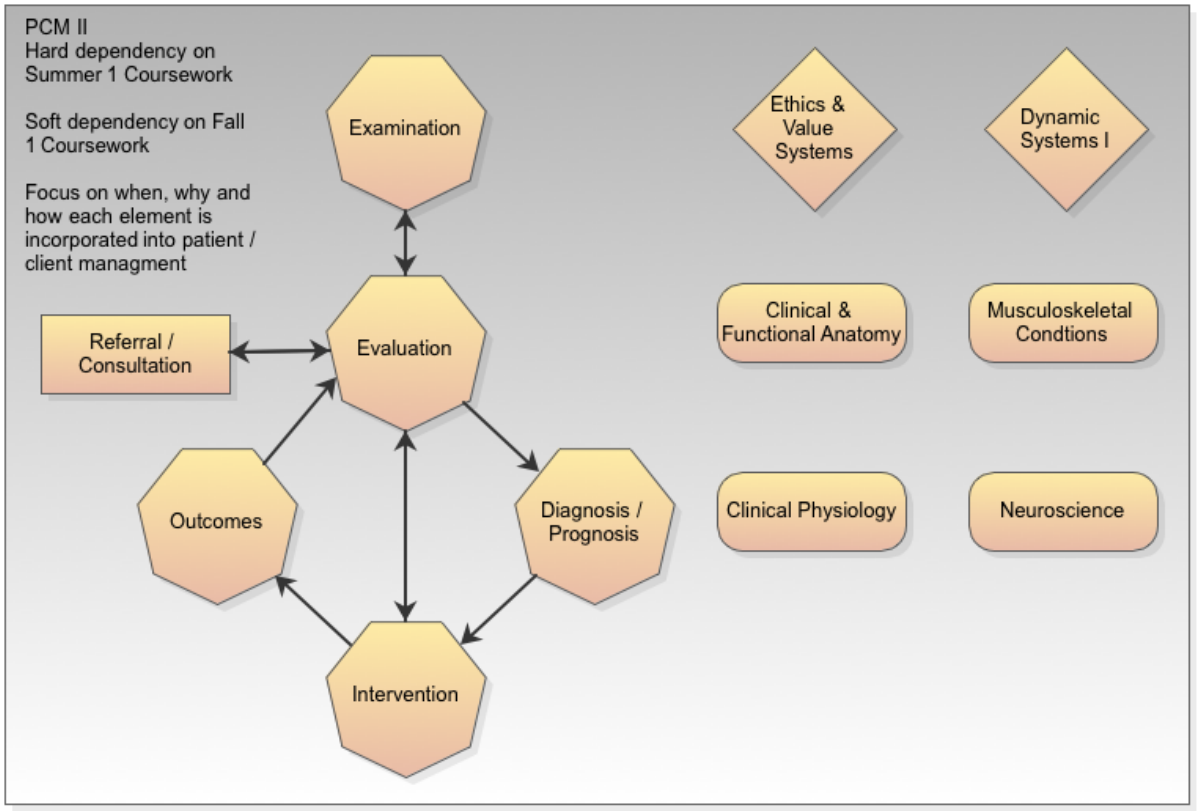


Figure 7. Patient / Client Management I



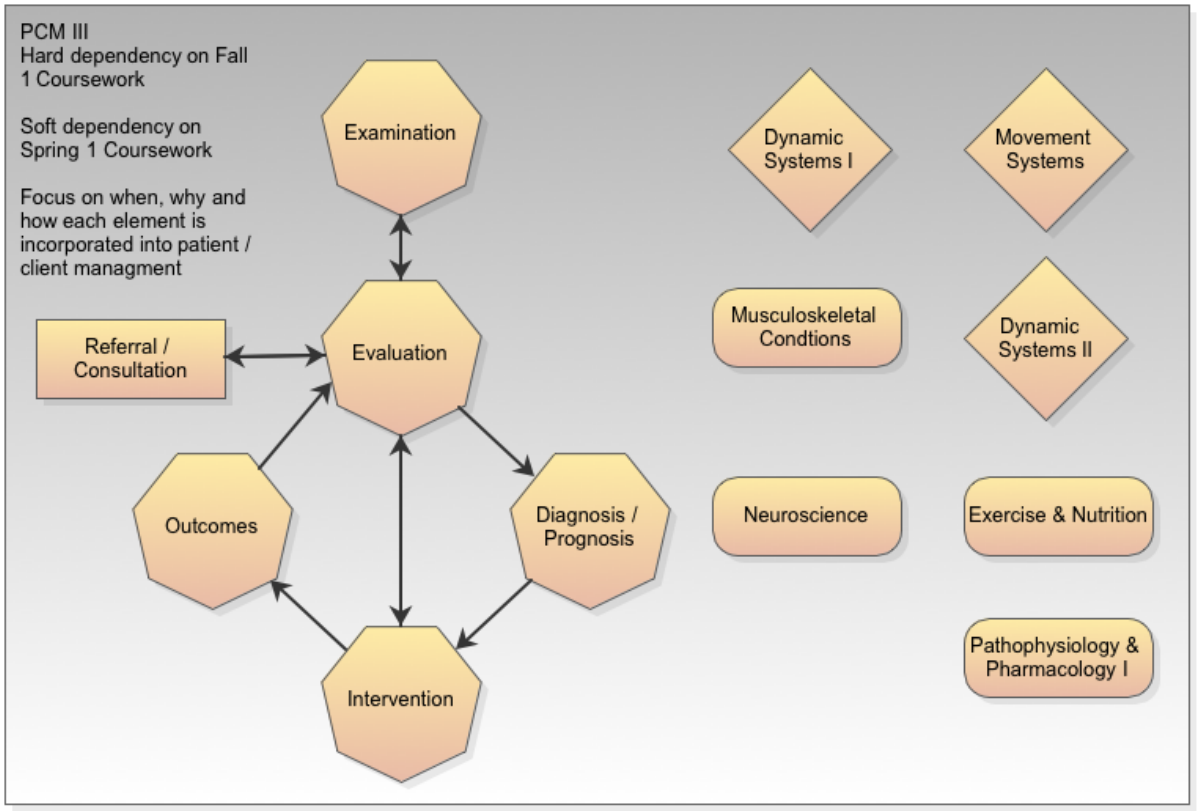
Scaffold of patient / client management and the explicit courses that form the hard (left column) and soft (right column) dependencies.

Figure 8. Patient / Client Management II



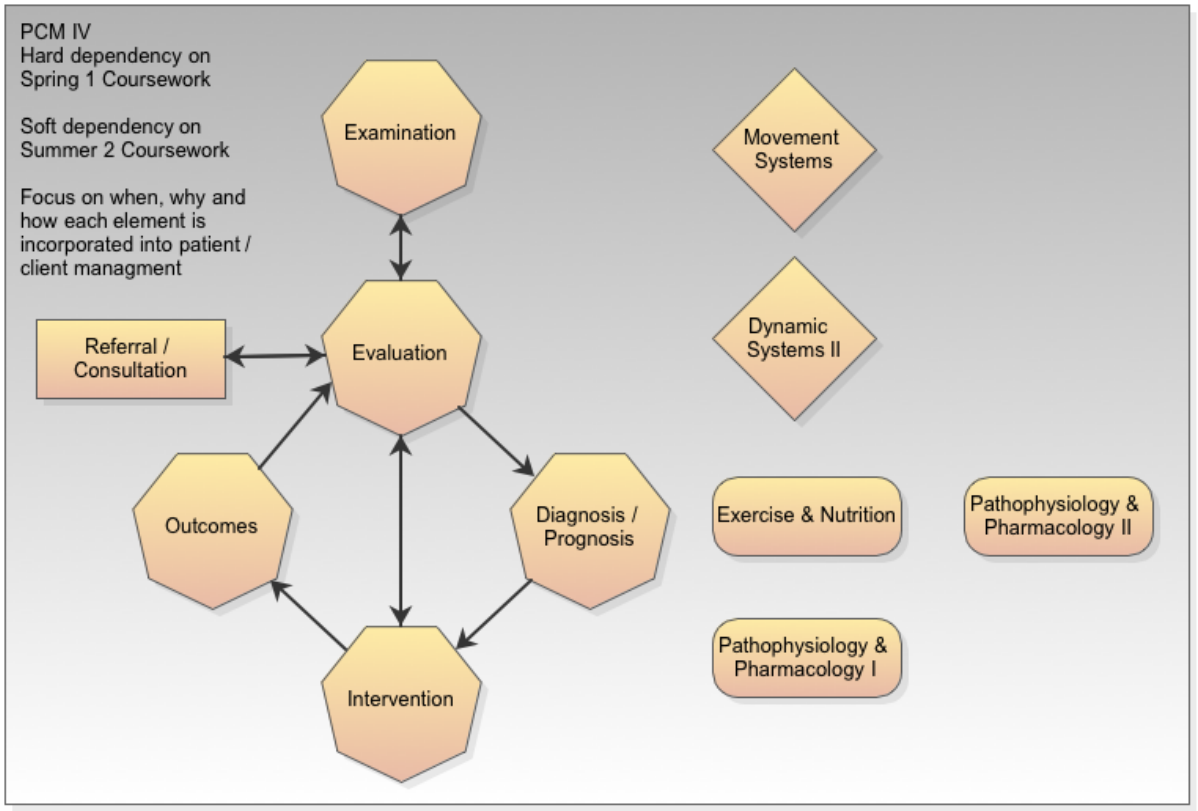
Scaffold of patient / client management and the explicit courses that form the hard (left column) and soft (right column) dependencies.

Figure 9. Patient / Client Management III



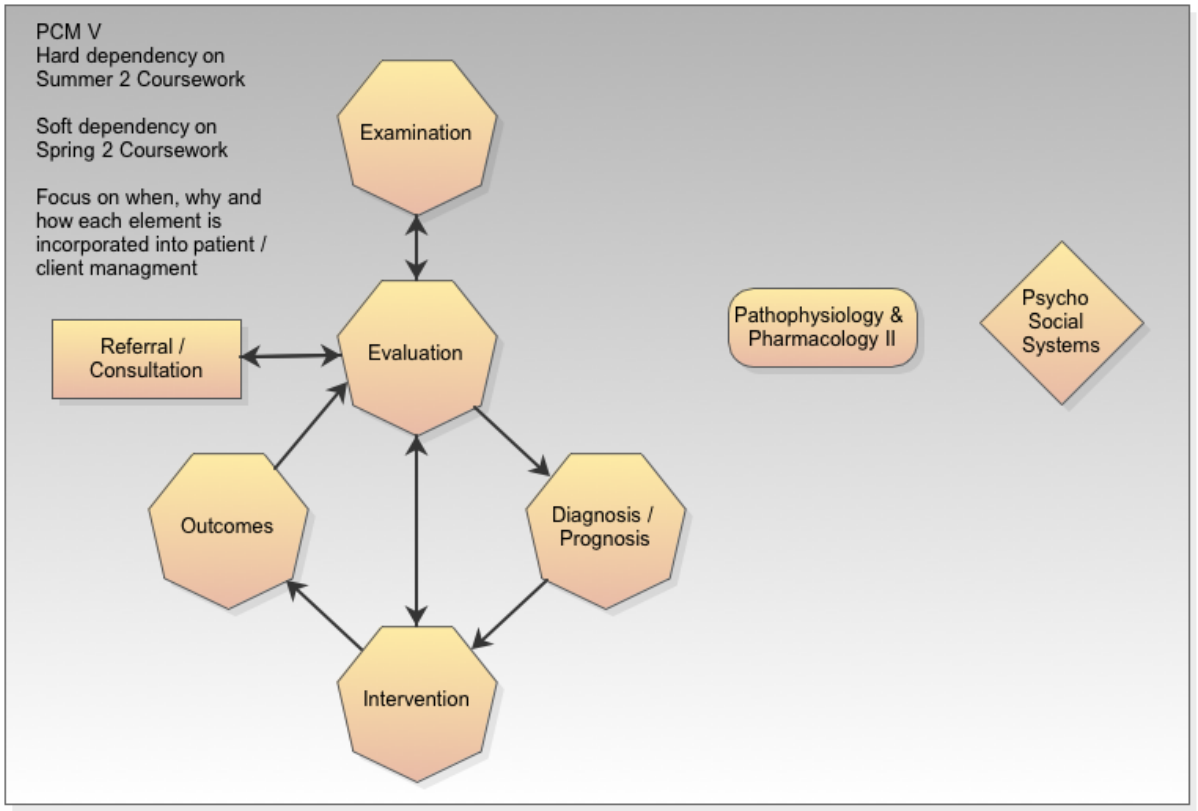
Scaffold of patient / client management and the explicit courses that form the hard (left column) and soft (right column) dependencies.

Figure 10. Patient / Client Management IV



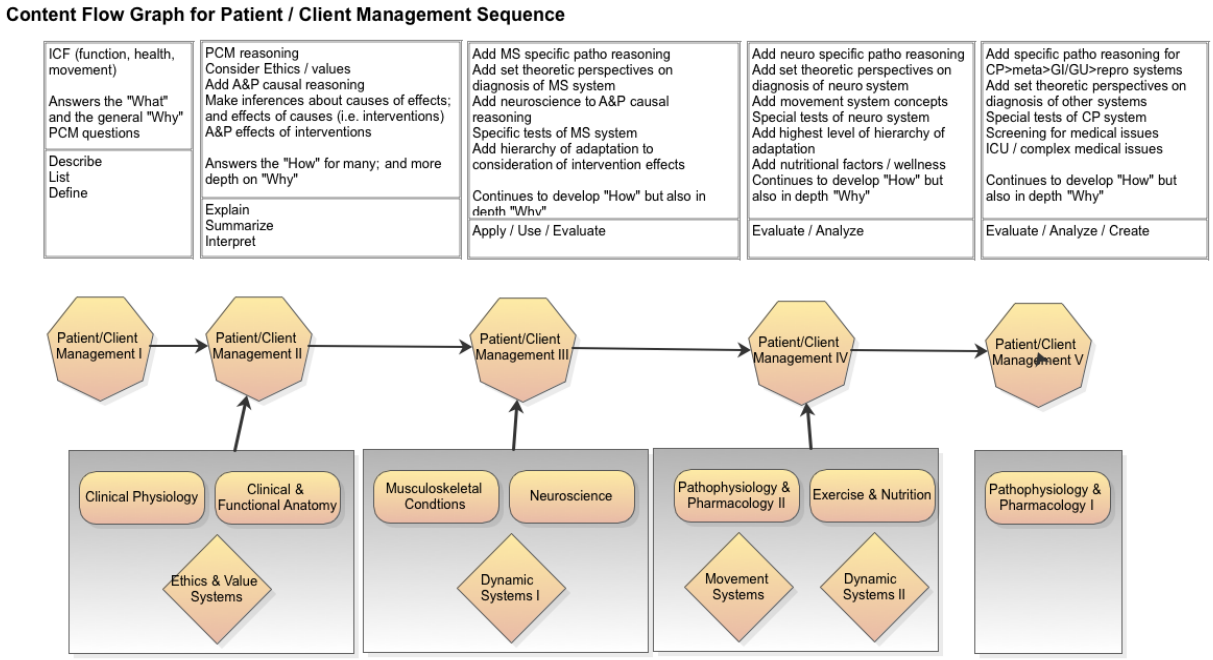
Scaffold of patient / client management and the explicit courses that form the hard (left column) and soft (right column) dependencies.

Figure 1. Patient / Client Management V



Scaffold of patient / client management and the explicit courses that form the hard (left column) and soft (right column) dependencies.

Figure 12. Content Flow



More insight into the content flow as the sequence snowballs. This graphic only includes the hard dependencies (pre requisite courses that precede the PCM course).

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[1] Graham C. Coming into Focus : The Need for a Conceptual Lens. JOPTE. 2015;29:5–12.
[2] Gwyer J, Hack L. The next big idea? JOPTE. 2015;29:4.
[3] http://www.apta.org/MovementSystem/, Retrieved on May 2, 2016
[4] https://en.wikipedia.org/wiki/System, Retrieved May 2, 2016
[5] Collins SM. Complex system approaches: could they enhance the relevance of clinical research? Phys Ther. 2005;85(12):1393-1394.
[6] Collins SM. On "Toward systems science in rehabilitation." Jette AM. Phys Ther. 2016;96:270-271. Phys Ther. 2016;96(5):743-743.
[7] Van Merriënboer, J. J. G., & Kirschner, P. A. (2012). *Ten steps to complex learning: A systematic approach to four-component instructional design*. book, Routledge.